

BIOL-1 Practice Test 04

Module 8: Cellular Genetics

Instructions: This practice test covers material from Module 8. Answer all questions to the best of your ability.

Part A: Multiple Choice (25 questions)

Choose the best answer for each question.

1. During which phase of interphase is DNA replicated?

- A) G₁
- B) S
- C) G₂
- D) M

2. What is the correct order of mitosis phases?

- A) Metaphase, Prophase, Anaphase, Telophase
- B) Prophase, Metaphase, Anaphase, Telophase
- C) Anaphase, Prophase, Metaphase, Telophase
- D) Telophase, Metaphase, Prophase, Anaphase

3. During which phase do chromosomes line up at the middle of the cell?

- A) Prophase
- B) Metaphase
- C) Anaphase
- D) Telophase

4. What separates during anaphase of mitosis?

- A) Homologous chromosomes
- B) Sister chromatids
- C) Nuclei
- D) Cell membranes

5. How many daughter cells are produced by mitosis?

- A) 1
- B) 2
- C) 3
- D) 4

6. Are the daughter cells of mitosis genetically identical to the parent cell?

- A) Yes
- B) No
- C) Only in plant cells
- D) Only in animal cells

7. In animal cells, cytokinesis occurs by:

- A) Cell plate formation
- B) Cleavage furrow
- C) Budding
- D) Binary fission

8. Diploid ($2n$) means a cell has:

- A) One set of chromosomes
- B) Two sets of chromosomes
- C) No chromosomes
- D) Half the normal number

9. How many chromosomes are in a human somatic (body) cell?

- A) 23
- B) 44

C) 46

D) 92

10. What is the purpose of meiosis?

A) Growth and repair

B) Making genetically identical cells

C) Making haploid gametes (sex cells)

D) Replacing dead cells

11. How many daughter cells does meiosis produce?

A) 1

B) 2

C) 3

D) 4

12. The daughter cells of meiosis are:

A) Diploid and identical

B) Diploid and unique

C) Haploid and identical

D) Haploid and unique

13. Crossing over occurs during:

A) Prophase I

B) Metaphase I

C) Anaphase II

D) Telophase II

14. What separates during Anaphase I of meiosis?

A) Sister chromatids

B) Homologous chromosomes

C) Individual genes

D) Cell membranes

15. Independent assortment occurs during:

- A) Prophase I
- B) Metaphase I
- C) Anaphase II
- D) S phase

16. In humans ($n = 23$), how many unique gamete combinations result from independent assortment alone?

- A) 23
- B) 46
- C) About 8 million
- D) About 70 trillion

17. Sister chromatids are:

- A) Chromosomes from different parents
- B) Two identical copies of one chromosome joined at the centromere
- C) Single-stranded DNA
- D) Found only in prokaryotes

18. Nondisjunction is when:

- A) DNA fails to replicate
- B) Chromosomes fail to separate correctly
- C) Crossing over doesn't happen
- D) A cell skips interphase

19. Down syndrome (trisomy 21) is caused by:

- A) A gene mutation
- B) Nondisjunction resulting in three copies of chromosome 21
- C) Too few chromosomes
- D) A missing X chromosome

20. A mutation converts a proto-oncogene into an oncogene. This could lead to:

- A) Cell death
- B) Slower cell division
- C) Uncontrolled cell division (cancer)
- D) No change

21. The tumor suppressor protein p53 functions to:

- A) Speed up cell division
- B) Stop cell division when DNA is damaged
- C) Build new chromosomes
- D) Perform crossing over

22. Which is NOT a source of genetic variation in sexual reproduction?

- A) Crossing over
- B) Independent assortment
- C) Mitosis
- D) Random fertilization

23. Homologous chromosomes are:

- A) Identical copies attached at the centromere
- B) Matching chromosomes, one from each parent
- C) Found only in bacteria
- D) The same as sister chromatids

24. What checkpoint verifies that all chromosomes are attached to spindle fibers before proceeding?

- A) G₁ checkpoint
- B) G₂ checkpoint
- C) M checkpoint (spindle checkpoint)
- D) S checkpoint

25. If a dog has 78 chromosomes in its body cells, how many chromosomes are in its sperm cells?

- A) 78
 - B) 39
 - C) 156
 - D) 19
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Part B: Fill in the Blank (10 questions)

- 26. The acronym for the phases of mitosis is _____ (four letters).
 - 27. DNA is copied during the _____ phase of interphase.
 - 28. Mitosis produces _____ (number) genetically identical daughter cells.
 - 29. Meiosis produces _____ (number) genetically unique daughter cells.
 - 30. Cells with two sets of chromosomes are called _____.
 - 31. Cells with one set of chromosomes are called _____.
 - 32. The exchange of DNA between homologous chromosomes is called _____.
 - 33. When chromosomes fail to separate properly during meiosis, it is called _____.
 - 34. A normal gene that promotes cell growth is called a _____; when mutated, it becomes an oncogene.
 - 35. In plant cells, cytokinesis occurs by forming a _____.
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Part C: Short Answer (4 questions)

- 36. Compare mitosis and meiosis. Include: number of divisions, number of daughter cells, ploidy, and purpose.

37. Name and describe the three sources of genetic variation in sexual reproduction.

38. What are cell cycle checkpoints? What can happen if they fail?

39. Explain why siblings from the same two parents look different from each other.

End of Practice Test 04